Gamma Ray Attenuation

Spandan Suthar, Zach Yockey, and Luke Childress

November 5, 2025

1 Introduction and Background

Radioactive materials emit a variety of particles as elements within them decay from unstable to stable isotopes, including gamma rays [3, 5]. These gamma rays can either interact with the matter and are prevented from passing through the material blocking their path, or they can pass through the material unaffected [3]. Interactions preventing gamma rays from passing directly through a material include the photoelectric effect, in which the gamma ray is exchanged for an electron; pair production, in which the gamma ray is exchanged for a positron-electron pair; and Compton scattering, in which the gamma ray imparts some energy to an electron and has its path redirected [3]. Studying the relationship between an object's properties (such as its length or its mass density) and its radiation-blocking efficacy allows for the safe use of radioactive materials in areas such as medical imaging, nuclear power generation, and further research involving radioactive substances. [5]

An effective measurement of a material's radiation-blocking efficacy is the mass attenuation coefficient $\mu = \frac{\alpha}{\rho}$, characterized by its role in the Beer-Lambert radiation intensity equation [3]:

$$I = I_0 e^{-\mu \cdot (\rho x)} = I_0 e^{-\alpha x} \tag{1}$$

This rule suggests that the intensity I of a gamma ray beam decreases exponentially relative to its initial intensity I_0, with a sensitivity to path length x through a material of mass density ρ given by the linear attenuation coefficient α. The dependence of α on the material's properties is typically removed by normalizing by ρ, giving μ. The intensity of radiation through a material is proportional to the number of gamma rays N passing through a material; (1) can be rewritten as [3, 5]:

$$N = N_0 e^{-\mu \cdot (\rho x)} = N_0 e^{-\alpha x} \tag{2}$$

This equation can be linearized to yield a form that will prove conducive to linear regression methods:

$$\ln(N) = \ln(N_0) - \alpha x \tag{3}$$

Linear interpolation is used when deriving accepted values of μ from public data. Given two points of a function (x_1, y_1) and (x_2, y_2), linear interpolation approximates the value of the function at x as:

$$y = y_1 + \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) \tag{4}$$

2 Methodology

An experimental setup is depicted in Figure 1. A radioactive sample of Cobalt-60 is placed in an appropriate, shielded container, which is placed under a scintillation detector. The detector signal is passed through an amplifier and into a multichannel analyzer (MCA), which provides a histogram of emission frequency versus channel number (a proxy for gamma ray energy), as in Figure 2. The experiment progresses by increasing the thickness of shielding material between the sample and the detector, thereby increasing the path length of the gamma rays x as in (1). For each plate of shielding material added, the MCA is allowed to collect counts from the detector for 10 minutes. Emission peaks are observed at different energies, yielding the datasets 3 and 2 of peak count N and shielding thickness x per emission peak observed, plate of shielding material added, and shielding material used. Note that Cobalt-60 exhibits two emission peaks (visible in Figure 2) and that two shielding materials are investigated: lead and aluminum.

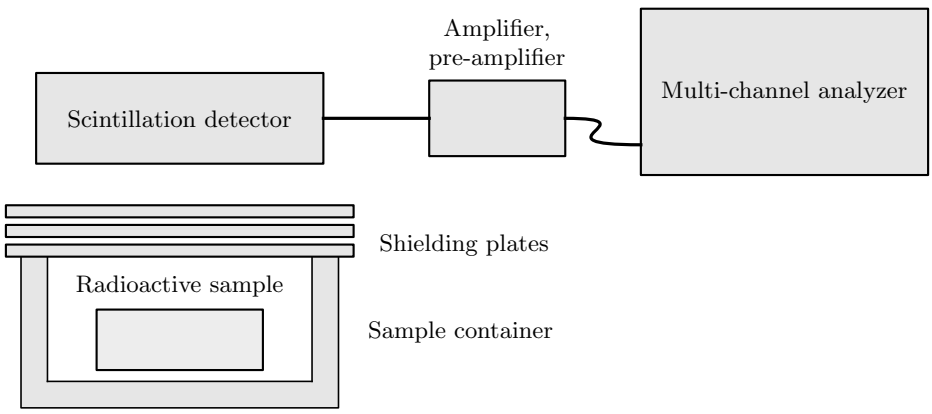

Figure 1: A schematic of the experimental apparatus. A radioactive sample is placed in a container shielding the experimentalist from radiation. Plates of shielding material are placed between the sample and a scintillation detector, which converts gamma rays into voltage signals. A pre-amplifier and amplifier pair adjust the signal to suitable parameters for the MCA: the voltage is set to an expected operation range and voltage pulses are shaped to be well-defined on an oscilloscope.

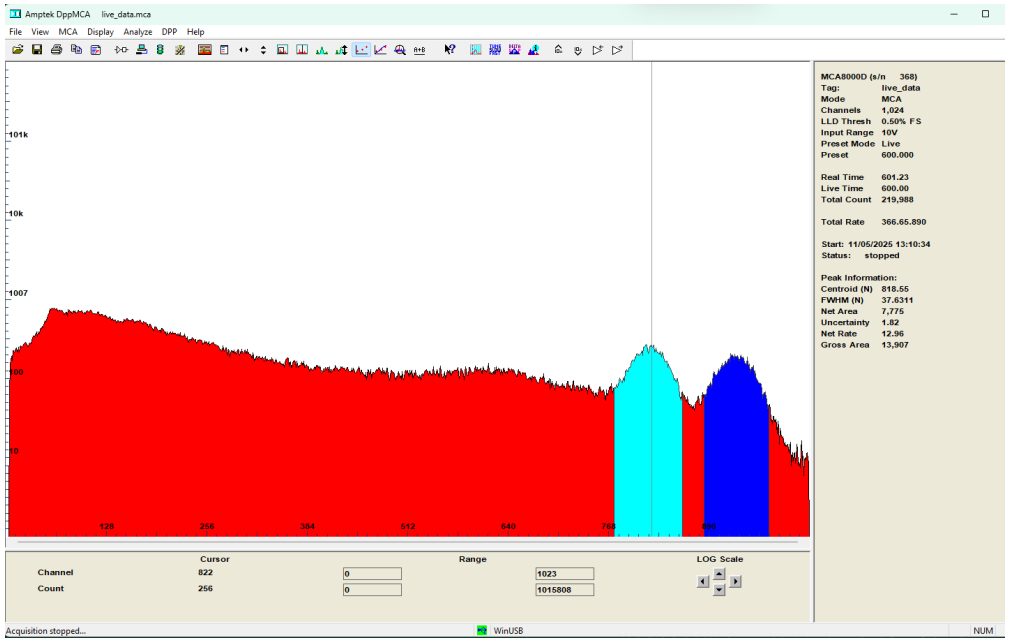

Figure 2: A spectrum of frequency versus channel number (energy proxy) as displayed by the MCA for Cobalt-60 with no shielding material. Note the two emission peaks of Cobalt-60, highlighted in light blue and dark blue.

3 Results

Peak 1 (counts)	Peak 2 (counts)	Thickness (cm)
225 ± 15	140 ± 12	0.00
211 ± 15	143 ± 12	1.30 ± 0.10
164 ± 13	152 ± 12	2.61 ± 0.20
137 ± 12	103 ± 10	4.0 ± 0.3
125 ± 11	90 ± 10	5.3 ± 0.4
111 ± 11	87 ± 9	6.6 ± 0.5
77 ± 9	56 ± 7	7.9 ± 0.6

Table 1: Gamma emission counts for aluminum shielding material at the peak 1 (1.1732 MeV) and peak 2 (1.3325 MeV) [4] of Cobalt-60 for different shielding material thicknesses. Counts were accrued in a 10-minute interval for each thickness.

Peak 1 (counts)	Peak 2 (counts)	Thickness (cm)
340 ± 18	261 ± 16	0.00
264 ± 16	187 ± 14	0.51 ± 0.10
162 ± 13	155 ± 12	1.05 ± 0.20
128 ± 11	108 ± 10	1.60 ± 0.30
76 ± 9	84 ± 9	2.10 ± 0.40
70 ± 8	53 ± 7	2.70 ± 0.50
54 ± 7	36 ± 6	3.20 ± 0.60

Table 2: Gamma emission counts for lead shielding material at the peak 1 (1.1732 MeV) and peak 2 (1.3325 MeV) of Cobalt-60 for different shielding material thicknesses. Counts were accrued in a 10-minute interval for each thickness.

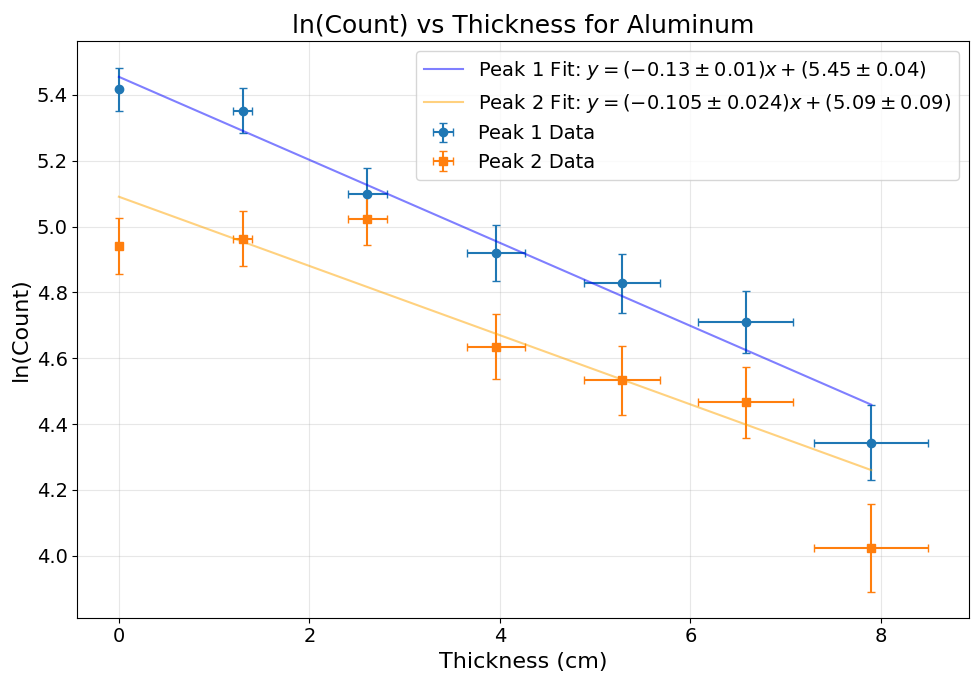

Figure 3: A plot of the natural logarithm of the peak count versus the thickness of the shielding material for aluminum, corresponding to Equation (3) and created with data derived from Table 3. Data for peak 1 is plotted in blue and data for peak 2 is plotted in orange. The slope corresponding to the least squares fit provides $-\alpha$, from which μ is derived. Origin not shown.

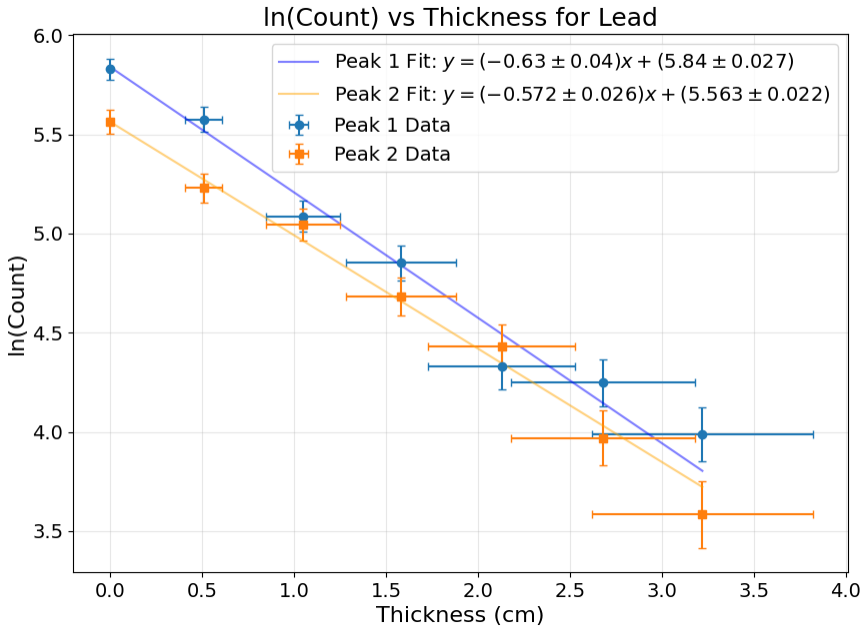

Figure 4: A plot of the natural logarithm of the peak count versus the thickness of the shielding material for lead, corresponding to Equation (3) and created with data derived from Table 2. Data for peak 1 is plotted in blue and data for peak 2 is plotted in orange. The slope corresponding to the least squares fit provides $-\alpha$, from which μ is derived. Origin not shown.

The data of tables 3 and 2 are plotted in Figures 3 and 4, respectively. Using a linear regression fit, slopes corresponding to $-\alpha$ are found, from which experimental values for the mass attenuation coefficient $\mu = \frac{\alpha}{\rho}$ for peaks 1 and 2 for aluminum and lead are derived. These values are reported in Table 3, along with discrepancies from the accepted values of μ obtained from the public NIST database via Equation (4) [1–3].

Experiment	Measured μ (m^2/kg)	Accepted μ (m^2/kg)	Discrepancy (σ)
1.1732 MeV Peak, Pb	0.0056 ± 0.0003	0.00625	2.2
1.3325 MeV Peak, Pb	0.00504 ± 0.00023	0.00566	2.7
1.1732 MeV Peak, Al	0.0047 ± 0.0004	0.00570	2.8
1.3325 MeV Peak, Al	0.0039 ± 0.0009	0.00533	1.6

Table 3: Experimental values of the mass attenuation coefficient μ for peaks 1 and 2 for aluminum and lead, respectively.

4 Discussion and Conclusion

Significant deviation from the accepted values of μ for both emission peaks and shielding materials is observed; for most trials, discrepancies are above 2σ. Primary causes for these discrepancies might include a lack of thorough shielding of the scintillation detector from external radiation, plates of shielding material that were not adequately large or aligned above the container cavity, and irregularities (local noise) in the peaks collected by the MCA. Despite an expected exponential

relationship between the number of gamma rays collected by the scintillation detector and the thickness of the shielding material, significant qualitative deviations of the aluminum data points from a linear trend are noted in Figure 3. A more thorough conduction of the experiment would involve larger plates to guarantee full coverage of the cavity and an average of the peak values per thickness and material to reduce the impact of peak irregularities.

References

[1] John H. Hubbell and S. M. Seltzer. X-ray mass attenuation coefficients - aluminum (z=13). Online; accessed 12 November 2025, 1995. Available at `https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z13.html`.

[2] John H. Hubbell and S. M. Seltzer. X-ray mass attenuation coefficients - lead (z=82). Online; accessed 12 November 2025, 1995. Available at `https://physics.nist.gov/PhysRefData/XrayMassCoef/ElemTab/z82.html`.

[3] John H. Hubbell and S. M. Seltzer. X-ray mass attenuation coefficients - section 2: Attenuation coefficient, cross section, methods. Online; accessed 12 November 2025, 1995. Available at `https://physics.nist.gov/PhysRefData/XrayMassCoef/chap2.html`.

[4] National Nuclear Data Center. Nudat: Nuclear data lookup from the national nuclear data center. `https://www.nndc.bnl.gov/nudat/`, 2025. Accessed: 2025-11-12.

[5] Hugh D Young and Roger A Freedman. *University Physics: with Modern Physics*. Harlow Pearson Education Limited, 15 edition, 2020.